Aabu Yousuf Raj

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Professional Profile

My strongest motivation as a researcher is the feeling that every solved problem reveals a deeper question. I am drawn to artificial intelligence, particularly machine learning, deep learning, and computer vision, because it connects rigorous theory with messy real-world patterns that are difficult to measure, interpret, and act upon. I look forward to pursuing research where careful experimentation and model design can produce useful real-world intelligence.

Education

Bachelor of Science (B.Sc.) in Computer Science — Sep 2022 to Feb 2026

BRAC University — *CGPA: 3.88/4.00* — Transcript

Graduated with Highest Distinction

Notable Courses: Advanced Algorithms, Software Engineering, Artificial Intelligence, Machine Learning, Data Science, Natural Language Processing II, Image Processing, Computer Vision, Neural Networks.

Academic Honors: Merit-Based Scholarship; Dean's List & Vice-Chancellor's List honoree for consistently outstanding academic performance.

Professional Experience

Student Tutor, BRAC University — *Teaching Assistance, Student Consultation, Assessment* — Feb 2025 to May 2025, Dhaka, Bangladesh

- Supported 40+ students through lab-based troubleshooting, one-on-one consultations, Q&A sessions, and targeted guidance for probation students, contributing to stronger understanding of course concepts and improved academic engagement.
- Graded assignments, maintained performance records, and organized digital course resources to streamline delivery of course material.

NIPPON AI Dojo Program — *Applied AI, AI Engineering, On-the-Job Training* — Jun 2026 to Present, Dhaka, Bangladesh

- Selected as one of 10 participants from BRAC University from a pool of 1,000+ applicants for an applied AI training program organized by Chowa Giken Corporation and AI Samurai Co. Ltd., with support from METI, Japan.
- Participating in industry-oriented on-the-job training focused on real-world AI development and Japanese engineering practices.

Academic Research

A Computer Vision Driven Ecosystem for Cattle Monitoring: Multi-Disease Classification with Severity Grading, Multi-View Individual Identification, and Weight Estimation — Feb 2025 to Present, Accepted for Publication

Undergraduate thesis *under the supervision of Dr. Jannatun Noor.*

Accepted for publication in ***Computers and Electronics in Agriculture (Elsevier, Q1)*** *as first author.*

- Proposed an RGB-based computer vision ecosystem for precision livestock farming, integrating multi-disease classification with clinically guided severity grading, multi-view individual identification, and reference-object-free body weight estimation using multi-view images and metadata.
- Designed and evaluated deep learning pipelines using attention-based hierarchical multi-task learning, YOLOv8-based cattle localization, ConvNeXt-based metric learning for multi-view cattle identification, and ensemble learning over fused RGB image and metadata features.
- Achieved 85.88% hierarchical disease-severity accuracy, 89.80% Rank-5 identification accuracy under cross-view domain shift, and 35.10 kg MAE in weight estimation, with Grad-CAM++ based explainability analysis supporting model interpretation for low-cost, low-shot, and non-invasive agricultural intelligence.

Academic Projects

Brain Tumor MRI Classification and Segmentation *Multi-task DL, Medical Imaging, U-Net* Github
Engineered a multi-task deep learning pipeline using U-Net, Attention U-Net, and ResUNet variants for simultaneous brain tumor segmentation and classification on MRI datasets. Improved model robustness through transfer learning, augmentation pipelines, and weighted loss functions to address class imbalance.

Real-Time Traffic Congestion Prediction in Dhaka *CNN, Image Classification, Urban AI* Github
Developed a CNN-based traffic intelligence system to classify congestion severity from street-level images in Dhaka's heterogeneous, lane-based urban environment. Collected, labeled, and preprocessed a custom street-imagery dataset to support data-driven traffic analysis and smart-city planning.

Token-Level NER with RNN Architectures *NLP, Sequence Labeling, BIO Tagging* Github
Conducted a comparative study of RNN, LSTM, GRU, and BiLSTM architectures for token-level Named Entity Recognition using the BIO tagging scheme on news datasets. Evaluated sequence-labeling performance to analyze the role of recurrent and bidirectional context modeling in information extraction.

Predicting Diseases from Symptom Profiles *Machine Learning, Healthcare AI, Classification* Github
Built a multi-class disease prediction pipeline from patient symptom profiles, applying feature engineering, Chi-Square based feature selection, and comparative classifier evaluation to identify effective models for symptom-driven clinical decision support.

Multimodal Music Clustering by VAE Variants *Unsupervised Learning, VAE, Multimodal Learning* Github
Built an unsupervised multimodal clustering system using VAE and Conditional VAE variants to group Bengali–English hybrid music in a shared latent space. Combined audio features, lyric embeddings, and genre conditioning, and evaluated cluster structure using visualization and clustering-quality metrics.

Technical Skills

Languages: Python, C/C++, Java, JavaScript, SQL, Bash, LaTeX
AI & ML Expertise: Computer Vision (CNNs, YOLO, ViT), Large Language Models (GPT, LLaMA), Transformers, RAG, Agentic Systems, Prompt Engineering, Fine-tuning, Custom Model Training
Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, Hugging Face, YOLO
Libraries: NumPy, Pandas, SciPy, Matplotlib, Seaborn, NLTK, Librosa, Albumentations
Tools: Git, GitHub, VS Code, Jupyter, Linux, Docker, CUDA, MySQL, Google Colab, Overleaf, CVAT, Roboflow

Leadership

Vice-President, BRAC University Marketing Association (BUMA) Oct 2024 to Feb 2026
Dhaka, Bangladesh
Led planning, promotion, sponsorship outreach, and logistics for university-wide technical events, seminars, programming contests, career workshops, and Capture The Flag (CTF) competitions, strengthening BUMA's campus and industry presence.

Test Scores

IELTS Academic: Overall 7.5 (CEFR C1)

References

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